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FEATURE ARTICLE

Conformationally flexible chiral supramolecular catalysts for enantioselective Diels–Alder reactions with anomalous *endo/exo* selectivitiesManabu Hatano^a and Kazuaki Ishihara^{*ab}

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The potential of supramolecular catalysts to realize anomalous regio- and/or stereoselectivity in organic synthesis is highly attractive. To date, there have been a few examples of non-polymeric and non-covalent chiral supramolecular catalysts that induce practical enantioselectivity. In this regard, a metal–organic framework (MOF) may be one of the most important techniques for constructing conformationally rigid supramolecular catalysts. However, it is not easy to use the MOF technique to fine-tune a much more precise cage in catalysts for anomalous purposes. To establish high catalytic activity with anomalous regio- and/or stereoselectivity, in principle, an artificial cage should be conformationally flexible, like an active pocket in an enzyme with an induced-fit function. In this feature article, we focus on the anomalous *endo/exo*-selective Diels–Alder reaction, and overview the development of the successive catalysts including our recent highly active, conformationally flexible, and chiral supramolecular catalysts. The evolution from ‘ready-made’ single-molecule catalysts to ‘tailor-made’ supramolecular catalysts could offer not only high enantioselectivity but also high anomalous *endo/exo*-selectivities due to substrate-specific characteristics, as with enzymes.

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1. Introduction

The Diels–Alder reaction is one of the most fundamental higher-ordered stereoselective reactions that involves the formation



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Manabu Hatano was born in Tokyo, Japan, in 1975, and received his PhD from the Tokyo Institute of Technology in 2003 under the direction of Professor Koichi Mikami. He was a JSPS Fellow under the Japanese Junior Scientists Program from 2000 to 2003. In 2003, he joined Professor Kazuaki Ishihara's group at Nagoya University as an assistant professor, and became associate professor in 2007. His research interests include asymmetric catalysis

with carbon–carbon bond-forming reactions with chiral acid–base salt catalysts, the chemistry of ate complexes for efficient catalytic asymmetric reactions, and the design of chiral self-assembled complexes toward supramolecular asymmetric catalysts.



Kazuaki Ishihara

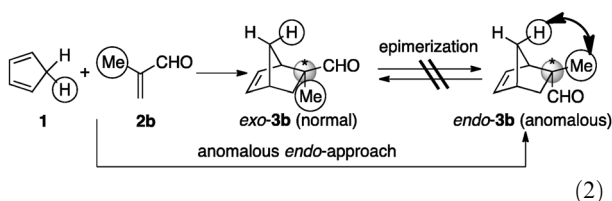
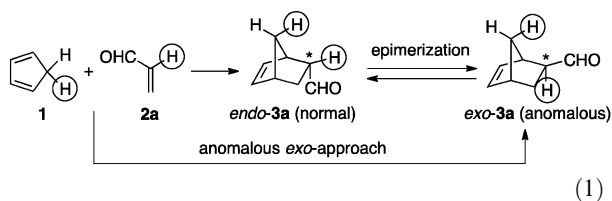
Kazuaki Ishihara was born in Aichi, Japan, in 1963, and received his PhD from Nagoya University in 1991 under the direction of Professor Hisashi Yamamoto. He had the opportunity to work under the direction of Professor Clayton H. Heathcock at the University of California, Berkeley, as a visiting graduate student for three months in 1988. He was a JSPS Fellow under the Japanese Junior Scientists Program from 1989 to 1991. After he completed his postdoctoral studies with

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of two carbon–carbon bonds.¹ By means of [4+2] cycloadditions, the efficient formation of a cyclohexane skeleton with four successive chiral carbon centers offers synthetic versatility, particularly in the total synthesis of many complex natural products.¹ In fact, in several studies to date, enantioselectivity in the Diels–Alder reaction has been successfully controlled by a variety of chiral catalysts or chiral auxiliaries in substrates. On the other hand, *endo/exo*-selectivity in the Diels–Alder reaction strongly depends on the substrates, based on Woodward and Hoffmann's frontier molecular orbital interactions and steric interactions between dienes and dienophiles *via* the formation of [4+2] pericyclic transition states under thermodynamic or photoreaction conditions (Fig. 1).²

Therefore, it is quite difficult to control not only enantioselectivity but also substrate-independent anomalous *endo/exo*-selectivity, since most conventional chiral catalysts can discriminate the enantiofaces of dienophiles, but not the approach of dienes (Fig. 2).¹ In fact, many combinations of dienes with dienophiles, particularly α,β -unsaturated carbonyl compounds, allow a well-known *endo*-rule that is based on second-order orbital interactions (Fig. 2a).² However, when steric interactions between dienes and dienophiles overcome the second-order orbital interactions in *endo*-transition states, less-familiar *exo*-adducts are often predominantly obtained against the *endo*-rule (Fig. 2b).²

For example, in the reaction between cyclopentadiene (**1**) and acrolein (**2a**), an *endo*-preference is observed with regard to second-order orbital interactions without significant steric interactions (eqn (1)). In sharp contrast, in the reaction between **1** and methacrolein (**2b**), an *exo*-preference is observed with regard to steric interaction between the methylene moiety of **1** and the methyl moiety of **2b** (eqn (2)).



Optically active *endo*-**3a** and *exo*-**3b** have been synthesized using many conventional chiral catalysts which can recognize dienophiles but not dienes. Moreover, thermodynamically

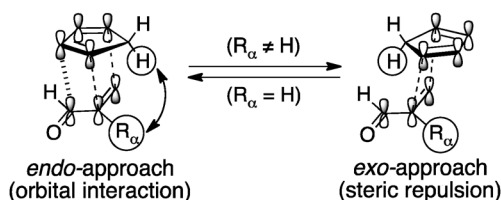


Fig. 1 Substrate-controlled *endo/exo*-selectivity.

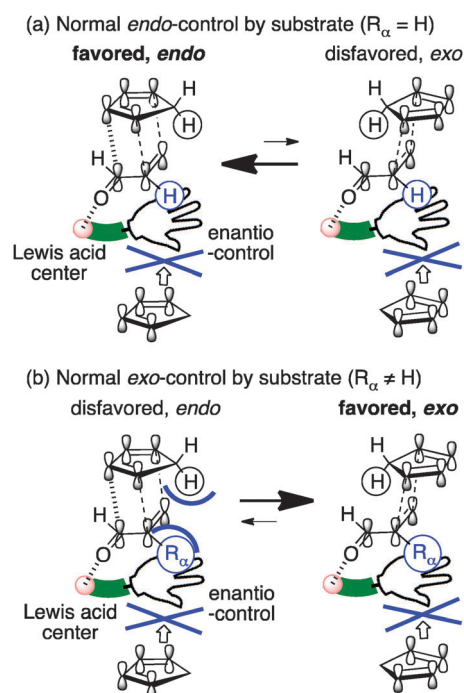


Fig. 2 Normal *endo/exo*-selectivity with conventional chiral catalysts.

more stable optically active *exo*-**3a**, which has a chiral tertiary carbon center at the 2-position, can also be generated by the epimerization of *endo*-**3a** (eqn (1)). In contrast, optically active

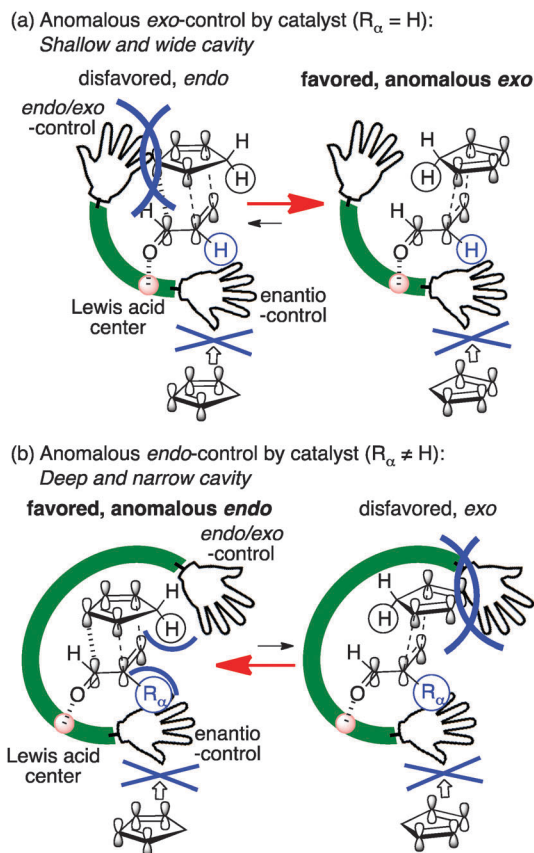


Fig. 3 Anomalous *endo/exo*-selectivity with specially designed chiral catalysts.

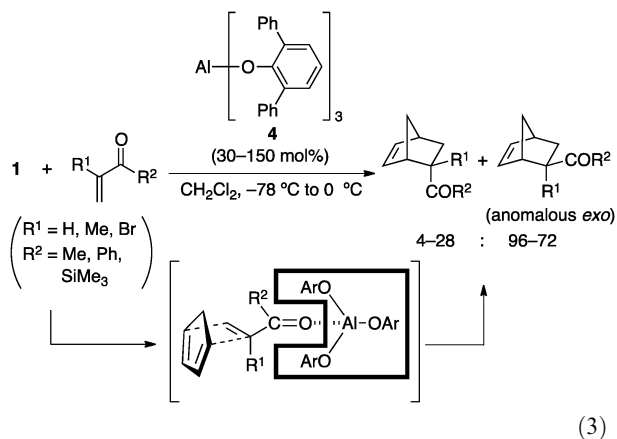
endo-**3b**, which has a chiral quaternary carbon center at the 2-position, cannot be generated by the epimerization of *exo*-**3b** or by other synthetic methods (eqn (2)).

To address the major unexplored subject in anomalous *endo/exo* control in the Diels–Alder reaction, catalysts must be able to accurately discriminate chiral transition state structures, by recognizing not only the *re/si*-face of dienophiles but also the *endo/exo*-approach of dienes (Fig. 3). First, catalysts must be able to discriminate the enantiofaces of dienophiles. Second, catalysts must be able to discriminate the *endo* and *exo* approach of dienes. Overall, the catalysts must discriminate both the diene and dienophile at the same time in transition states. For anomalous *exo*-control, the catalysts should have a shallow and wide cavity (Fig. 3a). On the other hand, for anomalous *endo*-control, the catalysts should have a deep and narrow cavity (Fig. 3b).

2. Anomalous *exo*-selective Diels–Alder reactions with achiral or chiral single-molecule catalysts

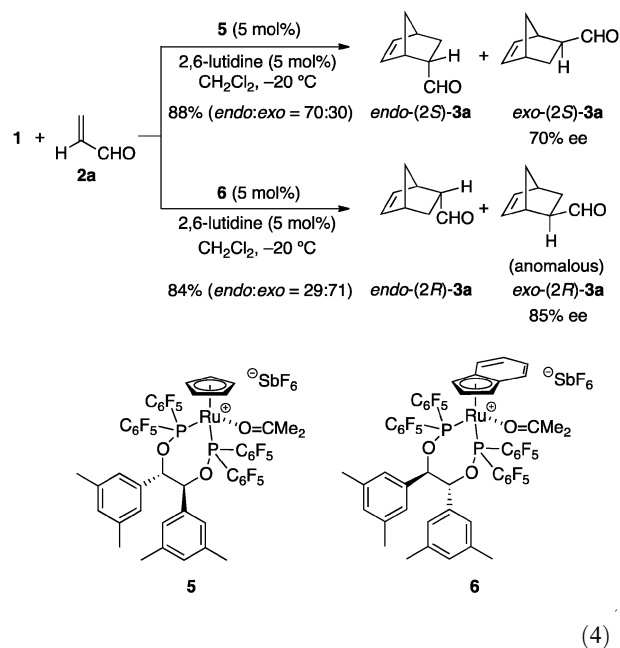
It should be easier to design promising catalysts for anomalous *exo*-control than for anomalous *endo*-control, since the external *exo*-approach can be realized when the internal *endo*-approach would be effectively prevented even with single-molecule catalysts with a relatively small structure (Fig. 3a). Thus, a deep and narrow cavity for both a diene and a dienophile is not necessary for anomalous *exo*-control.

In fact, relatively small single-molecule catalyst-induced anomalous *exo*-selective Diels–Alder reactions against the original *endo*-rule have been investigated by a few research groups. For example, Yamamoto *et al.* reported a molecular recognition approach by using a bulky aluminium(III) Lewis acid catalyst, ATPH (**4**), which provides an effective small carbonyl pocket, in a non-asymmetric manner for the first time (eqn (3)).³ Dienophiles are effectively shielded by complexation with the bulky achiral aluminium(III) catalyst, and the external *exo*-approach can be preferred (*endo* : *exo* = up to 4 : 96) while secondary interactions are significantly diminished.

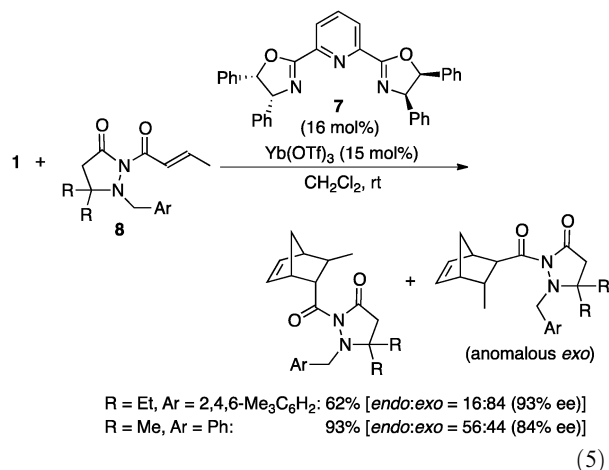


Later, Kündig *et al.* developed the first anomalous *exo*-selective, catalytic enantioselective Diels–Alder reaction between **1** and **2a** that usually favors the *endo* cycloadduct (eqn (4)).⁴ With cationic chiral indenyl ruthenium(II) catalyst (**6**), the reaction afforded anomalous *exo*-**3a** with 85% ee as major product

with an *endo* : *exo* ratio of 29 : 71. Interestingly, complete inversion of *endo* : *exo*-selectivity compared to that obtained with the cyclopentadienyl catalyst (**5**) was observed, which gave normal *endo*-**3a** as major product with an *endo* : *exo* ratio of 70 : 30. They concluded that the bulky indenyl arene ring would interfere with C(2)–H and C(3)–H of the approaching **1**.

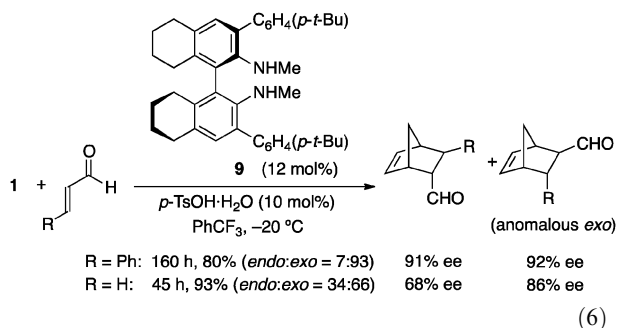


Sibi *et al.* developed the chiral Pybox (**7**)–ytterbium(III)-catalyzed anomalous *exo*-selective Diels–Alder reaction with a bulky achiral pyrazolidinone auxiliary in dienophiles (**8**) (eqn (5)).⁵ In this case, sterically hindered *N*-1 mesitylmethyl and *C*-5 ethyl substitutions in pyrazolidinone **8** were essential for realizing improved anomalous *exo*-selectivity (*endo* : *exo* = 16 : 84). For instance, the less-hindered *N*-1 benzyl- and *C*-5 methyl-substituted substrate led to a lower *endo* : *exo* ratio of 56 : 44.

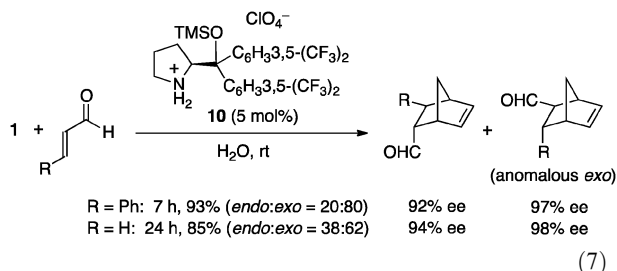


Maruoka *et al.* reported the chiral diamine salt-catalyzed anomalous *exo*-selective enantioselective Diels–Alder reaction (eqn (6)).⁶ The reaction of cinnamaldehyde in the presence of 12 mol% of chiral 3,3'-disubstituted *N,N'*-dimethyl *H*₈-binaphthyldiamine (**9**) and 10 mol% of *p*-TsOH·H₂O in

α,α,α -trifluorotoluene at $-20\text{ }^{\circ}\text{C}$ gave the corresponding anomalous *exo*-adduct with 92% ee (80% yield, *endo* : *exo* = 7 : 93). Moreover, the reaction of acrolein (**2a**) in the presence of the same catalyst gave the anomalous *exo*-**3a** with moderate *endo*/*exo*-selectivity (93% yield, *endo* : *exo* = 34 : 66, 86% ee for *exo*-**3a**). This is the first example of an organocatalytic anomalous *exo*-selective Diels–Alder reaction, although a prolonged reaction time was needed (40–160 h).



Hayashi *et al.* reported that a chiral salt of diarylprolinol silyl ether (**10**) induced anomalous *exo*-selectivity in the enantioselective Diels–Alder reaction between **1** and α,β -unsaturated aldehydes (eqn (7)).⁷ In particular, the reaction of cinnamaldehyde in the presence of 5 mol% of catalyst **10** in water at room temperature gave the corresponding anomalous *exo*-adduct with 97% ee (93% yield, *endo* : *exo* = 20 : 80). Moreover, the reaction of acrolein (**2a**) gave the anomalous *exo*-**3a** with moderate *endo*/*exo*-selectivity (*endo* : *exo* = 38 : 62, 98% ee for *exo*-**3a**).



3. Anomalous [4 + 2] and [2 + 2] reactions with conformationally rigid achiral or chiral supramolecular catalysts

Generally, for highly efficient and highly selective asymmetric carbon–carbon bond-forming reactions, chiral single-molecule catalysts are highly effective. To their advantage, they are small and easy to design. They are often effective for simple stereoselective reactions and show efficiency and generality for a wide range of substrates and reagents. However, they are usually not suitable for use in anomalous reactions to control unique multi stereoselectivity or substrate specificity. For this precise but important and difficult synthetic purpose, nano-size chiral ‘supramolecular catalysts’ should be attractive; they can eventually be quite large, but are easily tuned through the design of their small parts in advance.

Lehn developed the field of ‘supramolecular chemistry’, and explained its meaning in the first sentence of his paper in 1985;

‘Beyond molecular chemistry based on the covalent bond lies supramolecular chemistry based on molecular interactions—the associations of two or more chemical entities and the inter-molecular bond.’⁸ According to this understanding, the definition of ‘supramolecular catalysts’ is very simple and clear. Moreover, based on the novel design and fine-tuning of the constituent molecules, supramolecular catalysts can be expected to exhibit some special functions, unlike single-molecule catalysts.^{9–12}

Fujita *et al.* reported that a self-assembled coordination cage of palladium(II) (**11a**) could serve as a nano-sized molecular flask to promote the catalytic Diels–Alder reactions of aromatic hydrocarbons with *N*-cyclohexylmaleimide (**12**) (Fig. 4).¹⁰ Since the induced proximity of the reaction centers of two reactants in the confined cavity reduces the entropy of the Diels–Alder reactions, aromatic compounds that are otherwise unreactive, such as naphthalene, anthracene, and triphenylene, underwent unusual Diels–Alder reactions in a regio- and stereo-controlled manner under thermal or photo-irradiation conditions (Fig. 5).

Similar to Diels–Alder reactions (*i.e.*, [4 + 2] cycloaddition), [2 + 2] photoaddition reactions of polycyclic aromatic compounds and **12** were developed with achiral or chiral self-assembled pyridine–palladium cages (**11a** and **11b**) under irradiation with an ultra-high-pressure Hg lamp. Although the reaction scale was small (15 μmol) and a

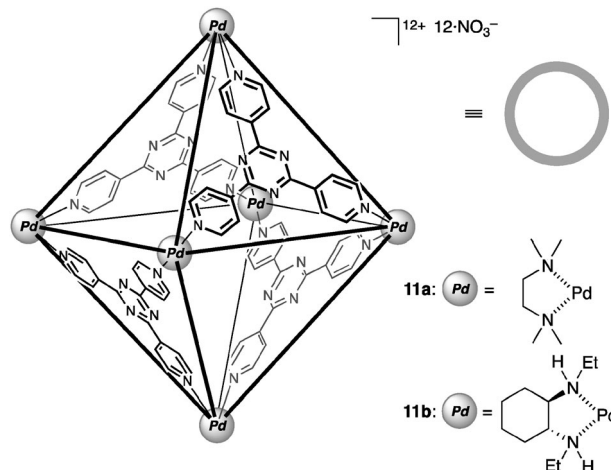


Fig. 4 Conformationally rigid self-assembled Pd₆ cages.

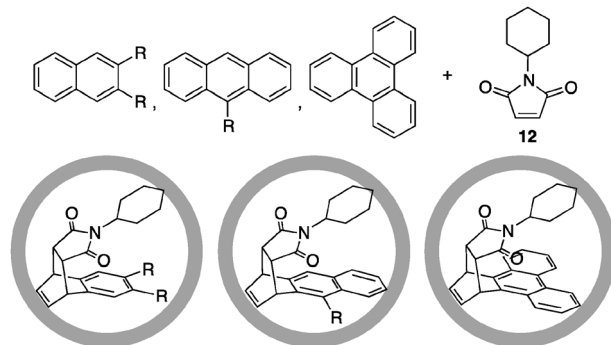
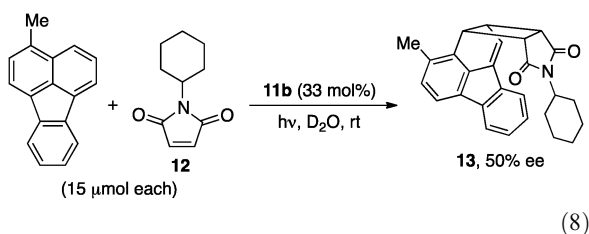


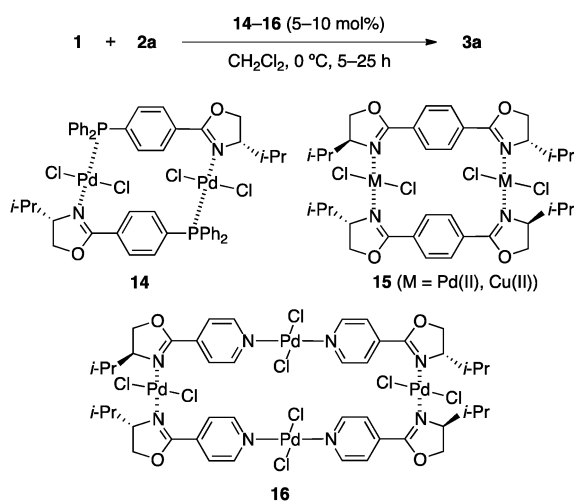
Fig. 5 Unusual Diels–Alder reaction in **11a**.

semi-equivalent amount of the chiral catalyst **11b** (33 mol% of the cage catalyst, 198 mol% for the palladium center) was used, moderate enantioselectivity (up to 50% ee) was observed in the corresponding adduct **13** (eqn (8)).^{10c} Basically, supramolecules with this type of metal framework cannot directly activate the substrate and/or reactant, and the catalytic process would not be so efficient as that in single molecule catalysts.



With regard to simple, rigid, and non-covalent metallo-supramolecular systems, we also developed chiral macrocyclic Pd(II) and Cu(II) complexes (**14–16**) based on *trans*-chelation of *n* : *n* metal-bidentate P,N- or N,N-ligands as shown in Scheme 1.¹³ These chiral supramolecular complexes were stable but did not exhibit high catalytic activity or anomalous stereoselectivity in the Diels–Alder reaction between **1** and **2a**, and the reactions proceeded very sluggishly without practical induction of enantioselectivity (0–86% yield, 0–38% ee).

In these stable cage and macrocyclic supramolecular catalysts, the metal centers might mainly play an important role for construction of the frame. However, due to the strong electron-donating ability of pyridines, the Lewis acidity would significantly decrease enough to activate the substrate in the catalytic reactions. Overall, these catalysts have conformationally rigid structures based on square-planar transition metal centers, and they don't have induced-fit properties that would cause them to adapt to dynamics in transition states.



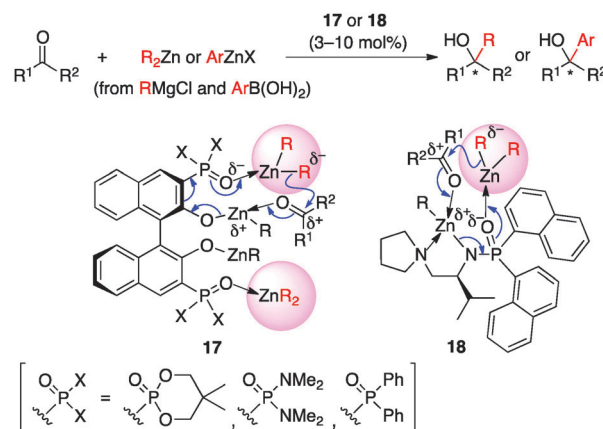
Scheme 1 Chiral macrocyclic Pd(II) and Cu(II) supramolecular catalysts based on *trans*-chelation of *n* : *n* metal-bidentate P,N- or N,N-ligands.

4. Anomalous *endo*-selective Diels–Alder reaction with conformationally flexible chiral supramolecular catalysts

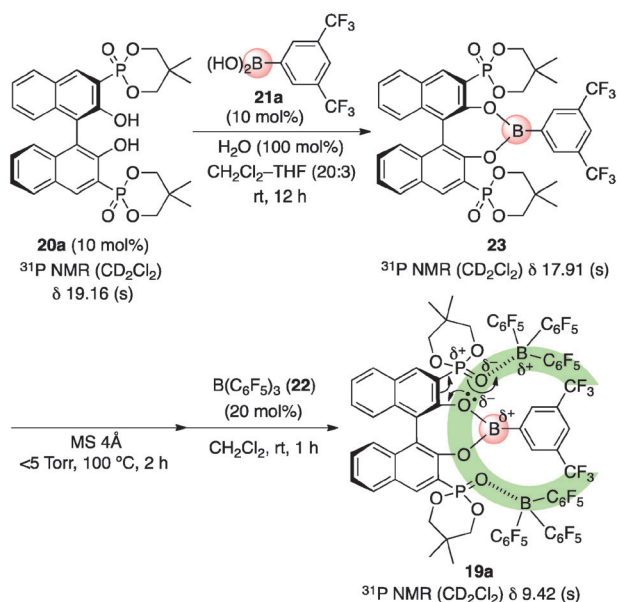
To establish high catalytic activity in unusual reaction pathways, the rational design of conformationally flexible chiral supramolecular catalysts, similar to enzymes, might be possible. In the course of our study on single-molecule catalysts, we have developed a catalytic enantioselective organozinc addition to ketones and aldehydes using acid–base conjugate Zn(II) catalysts (**17** and **18**) (Scheme 2).¹⁴ In these catalysts, acid–base remote activation of both the carbonyl compound and the organozinc reagent is important. In particular, the phosphoryl moieties would have good Lewis basicity to activate the electron-deficient organozinc center. We expected that this acid–base activation system, in particular with possible bulkiness based on the *C*₂-symmetric chiral BINOL moiety, might be applicable to the design of novel conformationally flexible chiral ‘supramolecular catalysts’.

Our advanced concept of conformationally flexible non-polymeric supramolecular catalysts is consistent with Lehn's original definition in that they include more than two molecules with non-covalent intermolecular bonds.⁸ Moreover, the ultimate use of our supramolecular catalysts is the asymmetric synthesis of unique target compounds that cannot be produced with the use of single-molecule catalysts. To achieve this goal, we have to extend the definition of traditional ‘supramolecular catalysts’ to more advanced molecules that possess enhanced catalytic activity and functionality for wonderful steric and electronic reasons in more than two molecules with non-covalent intermolecular bonds, unlike conventional chiral single-molecule catalysts.

In sharp contrast to previous anomalous *exo*-induced Diels–Alder reactions with α -non-substituted acroleins, to the best of our knowledge, there have been no reports on anomalous *endo*-induced catalytic enantioselective Diels–Alder reactions with α -substituted acroleins against the original *exo*-rule. In principle, for anomalous *endo*-control, a reaction intermediate should have a folding structure regardless of considerable steric repulsion between the reactants (Fig. 3b). Therefore, a deep and narrow cavity in the catalyst is necessary to hold both a diene and a dienophile together throughout the transition states.

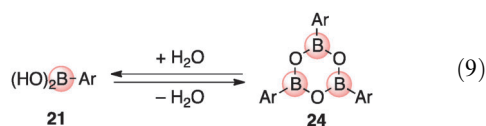


Scheme 2 Catalytic enantioselective organozinc addition to ketones using chiral Zn(II) catalysts with phosphoryl moieties.



Scheme 3 Preparation of a chiral supramolecular catalyst.

In such a situation, we developed a new type of conformationally flexible, highly active, chiral supramolecular catalyst based on well-designed single-molecule constituents.¹⁵ A chiral supramolecular catalyst (**19a**) was readily prepared in one pot *in situ* from three components, including 10 mol% of chiral (*R*)-3,3'-bis(5,5-dimethyl-2-oxido-1,3,2-dioxaphosphorinan-2-yl)-BINOL (**20a**)¹⁴ [BINOL = 1,1'-bi(2-naphthol)], 10 mol% of 3,5-bis(trifluoromethyl)phenylboronic acid (**21a**), and 20 mol% of tris(pentafluorophenyl)borane (**22**) (Scheme 3). Since the arylboronic acid (**21**) is usually allowed at hydrolysis equilibrium with the triarylboroxin (**24**) (eqn (9)), the addition of a small amount of water is necessary to provide the boron BINOLate intermediate (**23**).¹⁶ On the other hand, highly coordinative water molecules should be removed by MS 5 Å before the subsequent supramolecular formation of non-covalent bonds between **22** and **23**. Preliminary investigations by ^{31}P NMR analysis in CD_2Cl_2 showed a corresponding peak shift of two C_2 -symmetric phosphoryl moieties; 19.16 ppm for **20a**, 17.91 ppm for **23**, and 9.42 ppm for **19a**.



For the design of conformationally flexible supramolecular **19a**, intermolecular acid–base coordination bonds in the two $\text{P}=\text{O} \cdots \text{B}(\text{C}_6\text{F}_5)_3$ parts¹⁷ are critical. Compound **22** would act as a bulky functional group to make a chiral, narrow and deep cavity around the Lewis acidic boron center. Moreover, the strong electron-recipient ability of Lewis acid **22** would increase the Lewis acidity of the central boron through conjugate bonds (Scheme 3), which would take advantage of Lewis acid-assisted chiral Lewis acid (LLA) catalysts.¹⁸

Interestingly, once the substrate coordinates to the Lewis acid part of the original C_2 -symmetric supramolecular catalyst, the C_1 -symmetric intermediate would be generated

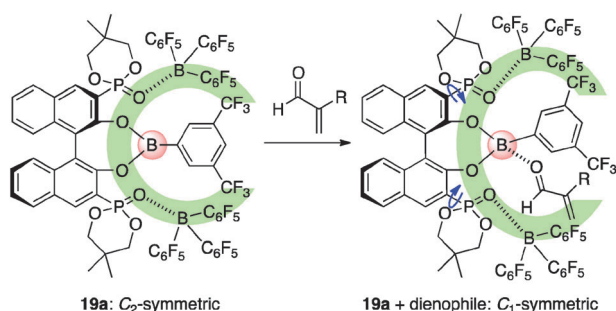


Fig. 6 A chiral cavity in conformationally flexible structures.

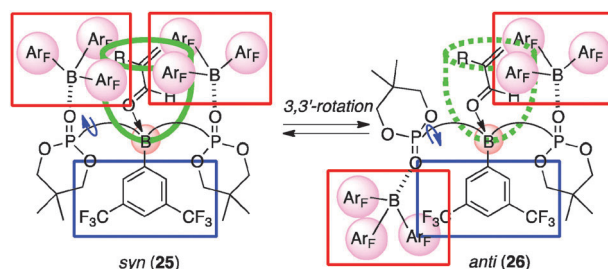
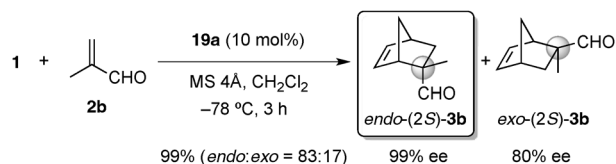


Fig. 7 *Syn*- and *anti*-conformations in the C_1 -symmetric intermediate.

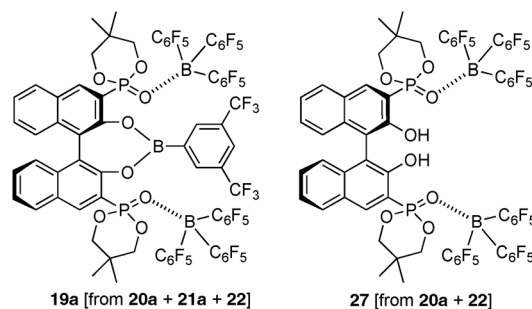
(Fig. 6). In this intermediate, the binaphthyl-3,3'-rotation of conformationally flexible bulky non-covalent parts of the phosphoryl- $\text{B}(\text{C}_6\text{F}_5)_3$ moiety would provide *syn*- and *anti*-conformations (**25** and **26**, respectively) (Fig. 7). In particular, in the *syn*-conformation **25**, a deep and narrow cavity, that would discriminate both a diene and a dienophile, is possible.

In the presence of the chiral supramolecular catalyst **19a** (10 mol%), the Diels–Alder reaction between **1** and **2b** was conducted in dichloromethane at -78°C for 3 h (Scheme 4). As a result, anomalous *endo*-(2*S*)-**3b** was obtained as a major

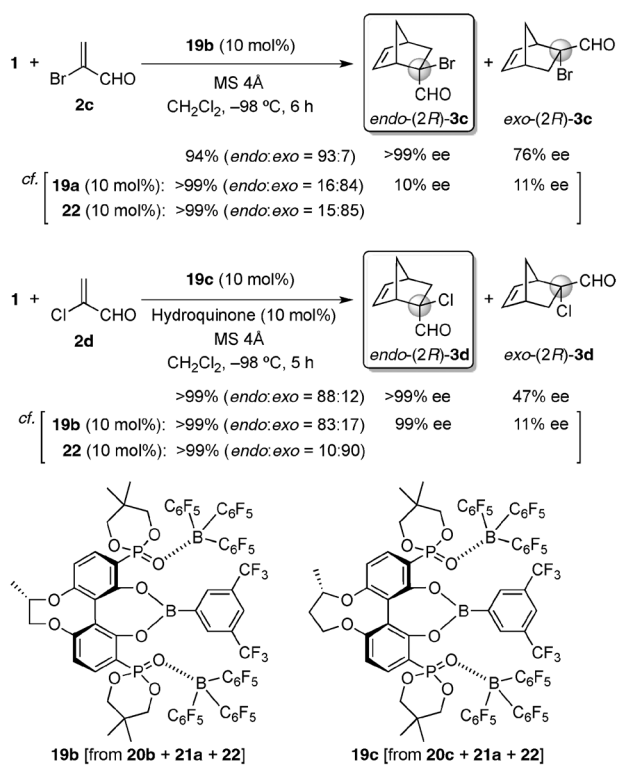


cf.

20a (10 mol%):	0%
21a (10 mol%):	>99% (<i>endo</i> : <i>exo</i> = 6:94)
22 (10 mol%):	93% (<i>endo</i> : <i>exo</i> = 12:88)
23 (10 mol%):	2%
27 (20 mol%):	0%



Scheme 4 Anomalous *endo*-selective enantioselective Diels–Alder reaction between **1** and **2b**.



Scheme 5 Anomalous *endo*-selective enantioselective Diels–Alder reaction between **1** and **2c** or **2d**.

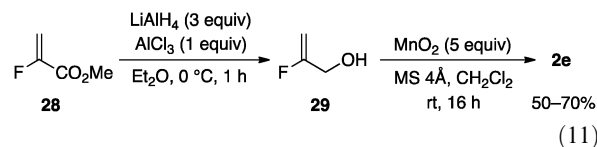
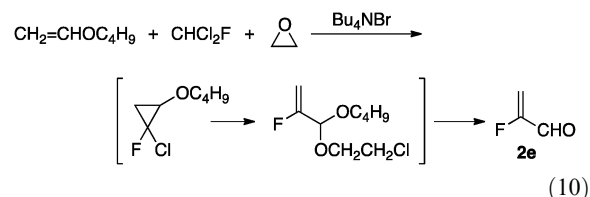
product (99% yield, *endo* : *exo* = 83 : 17) with excellent enantioselectivity (99% ee). In sharp contrast, **20a** and incomplete complexes **23** (i.e., [**20a** + **21a**]) and **27** (i.e., [**20a** + **22**]) showed almost no catalytic activity (0–2% yield). Moreover, **21a** and **22** gave the normal *exo*-**3b** as a major product (*endo* : *exo* = 6–12 : 94–88). For catalyst **27**, the phosphoryl moieties of **20a** would coordinate to **22**, and thus **22** as a Lewis acid would be deactivated. For catalyst **23**, the corresponding boron BINOLate would not be reactive, since it lacks conjugated activation by Lewis acid **22**. Therefore, the catalytic activity of **19a** decreased even in the absence of either **21a** or **22**.

To explore anomalous *endo*-selective Diels–Alder reactions, we examined the reactions between **1** and α -haloacroleins, which usually provided *exo*-adducts as major products (see the **22**-catalysis in Scheme 5). Electron-deficient α -haloacroleins are extremely reactive, and thus their enantioselective Diels–Alder reactions have been especially limited.^{19–21} Moreover, the reports were on substrate-dependent *exo*-selective reactions. Corey *et al.* reported pioneering *exo*-selective examples with both α -bromoacrolein (**2c**)¹⁹ and α -chloroacrolein (**2d**),²⁰ and other research groups later reported *exo*-selective and enantioselective catalysis with **2c**.²¹ However, there have been no catalytic asymmetric examples of the Diels–Alder reactions of α -fluoroacrolein (**2e**).

In the reaction between **1** and **2c** in dichloromethane at –98 °C for 6 h, supramolecular catalyst **19a** was unexpectedly ineffective, and *exo*-**3c** was obtained as a major product with low enantioselectivity (>99% yield, *endo* : *exo* = 16 : 84, 10–11% ee). However, after optimization of the chiral biaryl skeleton, we found that another supramolecular catalyst **19b** with chiral biphenol **20b** in place of chiral binaphthol **20a** was

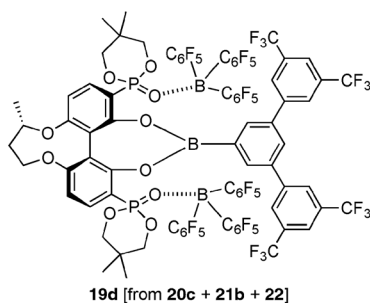
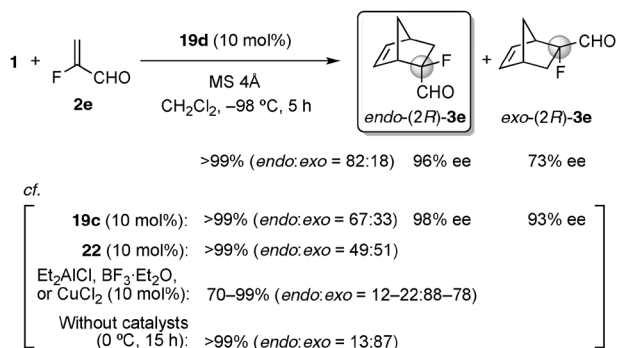
extremely effective, and the anomalous *endo*-selectivity was dramatically improved (94% yield, *endo* : *exo* = 93 : 7) with excellent enantioselectivity for *endo*-(2*R*)-**3c** (>99% ee). Furthermore, in the reaction between **1** and **2d** in dichloromethane at –98 °C for 5 h in the presence of hydroquinone (10 mol%) as a polymerization inhibitor, after another fine-tuning for the chiral biaryl skeleton, supramolecular catalyst **19c** provided anomalous *endo*-selectivity, and *endo*-(2*R*)-**3d** was obtained as a major product (>99% yield, *endo* : *exo* = 88 : 12) with excellent enantioselectivity (>99% ee). Supramolecular catalyst **19b** was slightly less effective for **2d** than **19c** in increasing the anomalous *endo*-selectivity.

Next, α -fluoroacrolein (**2e**) was examined. Although a method for the preparation of **2e** has been reported, as shown in eqn (10), gaseous dichlorofluoromethane (bp 8.9 °C) and ethylene oxide (bp 10.7 °C) are not convenient to use.²² Alternatively, we used commercially available methyl 2-fluoroacrylate (**28**) as a starting material, and 2-fluoro-2-propen-1-ol (**29**) was synthesized by the reported procedure with AlH_3 that was generated *in situ* from LiAlH_4 and AlCl_3 .²³ Without further purification of the alcohol **29**, subsequent oxidation with MnO_2 selectively provided **2e** (eqn (11)). Compound **2e** is volatile (bp 71–72 °C),²² and **2e** was obtained in ca. 50–70% yield as a solution in dichloromethane (0.10–0.30 M) after filtration of the resultant mixture and distillation.



Under thermodynamic conditions without catalysts, *exo*-product **3e** was obtained predominantly (>99% yield, *endo* : *exo* = 13 : 87) (Scheme 6). An *exo*-preference was also observed when representative Lewis acid catalysts, such as Et_2AlCl , $\text{BF}_3 \cdot \text{Et}_2\text{O}$, and CuCl_2 , were used (*endo* : *exo* = 12–22 : 88–78), while a 1 : 1 *endo* : *exo* ratio was observed when bulky **22** was used. On the other hand, anomalous *endo*-(2*R*)-**3e** was obtained as a major product (>99% yield, *endo* : *exo* = 82 : 18) with high enantioselectivity (96% ee) when bulky **19d**, which was derived from a chiral biphenol **20c**, 3,5-bis[3,5-bis(trifluoromethyl)phenyl]phenylboronic acid (**21b**) and **22**, was used as an optimal supramolecular catalyst. In this case, precedent **19c**, which was a good catalyst for **2d**, was not very effective for an anomalous *endo*-preference although high enantioselectivity was observed for *endo/exo*-**3e**.

Overall, as with an enzymatic methodology, fine-tuning of the conformationally flexible supramolecular catalysts for each α -haloacrolein was essential for establishing anomalous *endo*-selectivity as well as excellent enantioselectivity. As the halogen in α -haloacrolein became smaller, larger components at the central aryl borane moiety and the biphenyl moiety were



Scheme 6 Anomalous *endo*-selective enantioselective Diels–Alder reaction between **1** and **2e**.

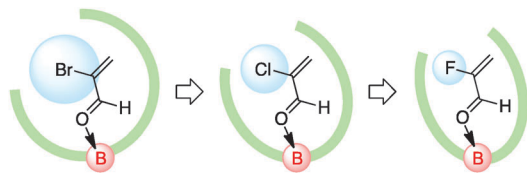


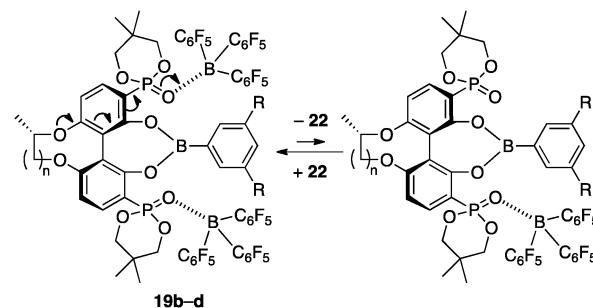
Fig. 8 Tuning the size of a suitable cavity for each substrate.

effective. The bulkiness may directly or indirectly create a smaller cavity that could suitably recognize a smaller substrate (Fig. 8).

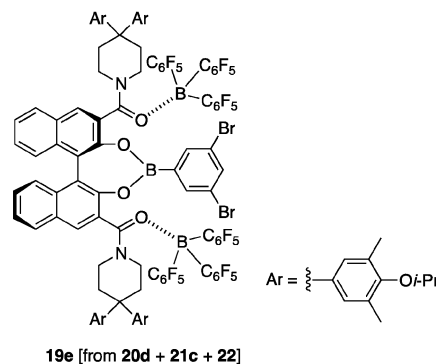
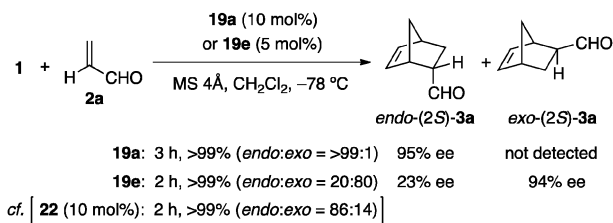
Moreover, it is not fully clear why the anomalous *endo*-diastereoselectivity of **3c–e** was significantly improved when biphenyl catalysts **19b–d** were used in place of binaphthyl catalyst **19a**. However, as one possible explanation, there might be a slight difference in the dihedral angle of the binaphthyl or biphenyl skeleton. As another possible explanation, the electron-donating ability of the 6,6'-ether moieties in **19b–d** (i.e., –OCH(CH₃)CH₂O– or –OCH(CH₃)CH₂CH₂O–), through a resonance effect in the conjugate system, might induce a stronger intermolecular acid–base coordination of non-covalent P=O...B(C₆F₅)₃ (Scheme 7). This stabilization of the supramolecular catalysts might reduce the adventitious dissociation of achiral **22**. Consequently, normal Diels–Alder reactions with extremely low enantioselectivity by incomplete supramolecular catalysts and/or **22** would be prevented when reactive **2c–e** were used in place of much less reactive **2b**.

5. Anomalous *exo*-selective Diels–Alder reaction with a conformationally flexible chiral supramolecular catalyst

Next, we examined the reaction with less-reactive acrolein (**2a**) in place of α -haloacroleins (Scheme 8). Generally, the reaction of **1** and **2a** was *endo*-selective under substrate-control



Scheme 7 A resonance effect in chiral biphenol catalysts.



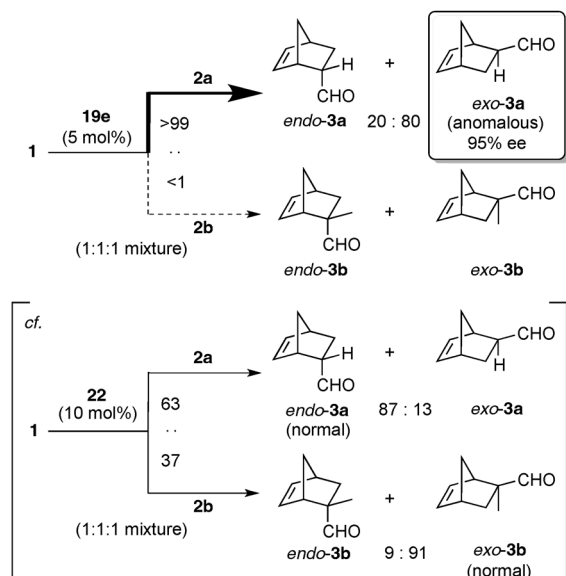
Scheme 8 Extremely *endo*-enhanced or anomalous *exo*-selective enantioselective Diels–Alder reaction between **1** and **2a**.

(e.g., *endo* : *exo* = 86 : 14 by **22**-catalysis). As expected, the reaction with supramolecular catalyst **19a** (10 mol%) provided *endo*-(2*S*)-**3a** in >99% yield with perfect *endo*-selectivity (*endo* : *exo* = >99 : 1) and high enantioselectivity (95% ee).

Moreover, we developed another supramolecular catalyst with amido moieties in place of phosphoryl moieties. As with supramolecular catalysts **19a–d** with phosphoryl moieties, **19e** (5 mol%) was prepared *in situ* from chiral 3,3'-(dicarbamoyl)-binaphthol (**20d**), (3,5-dibromophenyl)boronic acid (**21c**), and **22**. Very interestingly, this catalyst was highly effective for the anomalous *exo*-selective Diels–Alder reaction of **2a** with high enantioselectivities (94% ee for *exo*-(2*S*)-**3a**, *endo* : *exo* = 20 : 80).

6. Molecular recognition by the chiral supramolecular catalyst

To further investigate the function of conformationally flexible supramolecular catalysts, a preliminary examination of molecular recognition under competitive Diels–Alder reaction conditions was performed. For a 1 : 1 : 1 equimolar mixture of **1**, **2a**, and **2b**, *exo*-inducing supramolecular catalyst **19e** (5 mol%) promoted the reaction of **2a** exclusively (**3a** : **3b** = >99 : <1), and anomalous *exo*-(2*S*)-**3a** was obtained as a major product (*endo* : *exo*-**3a** = 20 : 80, 95% ee for *exo*-(2*S*)-**3a**) (Scheme 9). In sharp contrast,



Scheme 9 Molecular recognition under competitive Diels–Alder reaction conditions.

achiral Lewis acid catalyst **22** (10 mol%) gave a mixture of *endo*-**3a** and *exo*-**3b** as major products with low substrate-selectivity (**3a** : **3b** = 63 : 37) and normal *endo*/*exo*-selectivity (*endo* : *exo*-**3a** = 87 : 13, *endo* : *exo*-**3b** = 9 : 91). This result suggests that the supramolecular catalyst may have some induced-fit functions to adapt to a specific substrate.

7. Mechanistic study through possible intermediates and possible transition states

We cannot fully explain why anomalous *endo*-selectivity was observed in **19a–d**-catalysis for α -substituted acroleins while anomalous *exo*-selectivity was observed in **19e**-catalysis for acrolein. Nevertheless, as a working model, we turn our attention to mechanistic aspects involving the geometry of the possible supramolecular catalyst **19a**. A chiral, narrow, and deep cavity is assumed due to the six bulky C_6F_5 moieties and the binaphthyl moiety (Fig. 7). The two non-covalent $P=O \cdots B(C_6F_5)_3$ moieties could have a *syn*-conformation (**30**) on the one hand and an *anti*-conformation (**31**) on the other hand, and a preference for *syn*-**30** is clearly illustrated by theoretical calculations for a **2b–19a** complex with the B3LYP/6-31G* method (Fig. 9).^{24–26} Interestingly, *syn*-**30** is more stable than *anti*-**31** by 3.86 kcal mol^{−1}. In *anti*-**31**, significant steric repulsion would be observed among the C_6F_5 moieties and the 3,5-(CF_3)₂ C_6H_3 moiety, while *syn*-**30**, which lacks significant steric repulsion, is more favored.

In *syn*-**30**, the formyl moiety of **2b** with a favored *s-trans* geometry was doubly coordinated with the $B-O(Naph)$ moiety at the $C(=O)H$ and $C(=O)H$ parts (Fig. 9). Among the possible transition states, a *si*-face attack would be disfavored due to enantio-face control by a C_6F_5 group. In a possible transition state (TS)-**32**, an *endo*-approach inside the cavity via a *re*-face attack would be relevant, while an *exo*-approach via a *re*-face attack (TS-**33**) would be unlikely because of the bulkiness of another C_6F_5 group (Fig. 10).

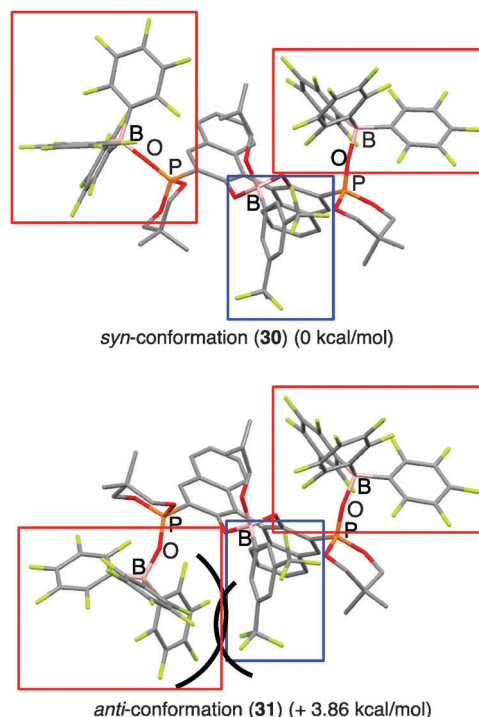


Fig. 9 Theoretical calculations using Gaussian03 with the B3LYP/6-31G* basis set.

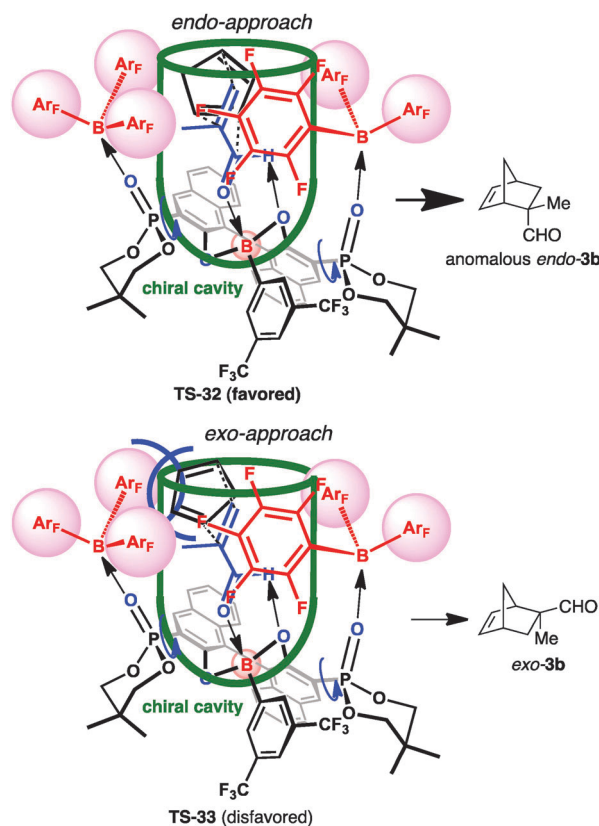


Fig. 10 Possible transition states for anomalous *endo*-selectivity.

Similar to the case of supramolecular **19a** with phosphoryl moieties, possible transition states (TS-**34** and TS-**35**) for supramolecular **19e** with amide moieties are shown in Fig. 11.

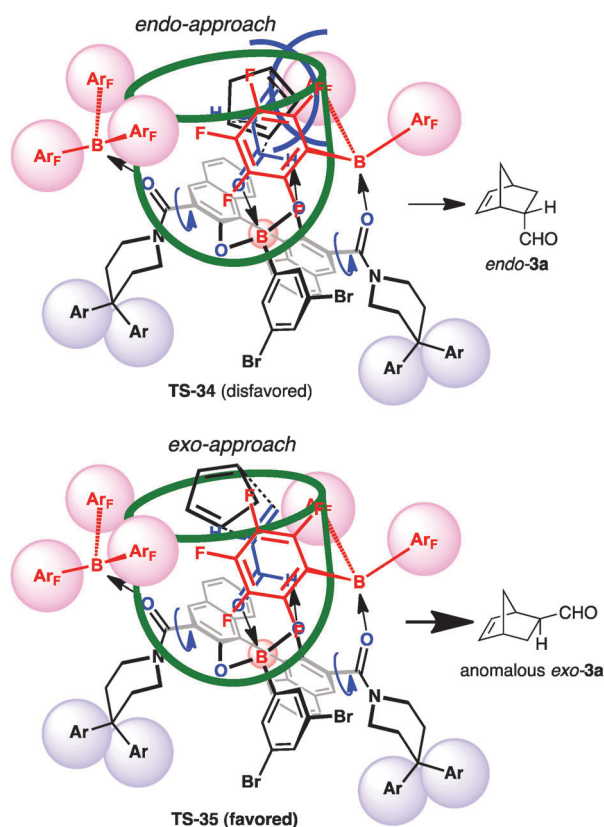


Fig. 11 Possible transition states for anomalous *exo*-selectivity.

Unlike the pseudo-tetrahedral phosphorous structure, the amide has a less-hindered planar structure, and the non-covalent amide- $\text{B}(\text{C}_6\text{F}_5)_3$ moiety may turn outside the **1-2b-19e** complex in the transition states. Therefore, a shallow and wide cavity would be generated, and this would promote the anomalous *exo*-approach without significant steric repulsion via TS-35.

8. Conclusions

The anomalous *endo/exo*-selective Diels-Alder reaction is one of the most interesting and difficult subjects in organic synthesis, particularly with catalytic asymmetric approach, since the *endo/exo*-selectivity is usually dependent on the substrates used. To date, some pioneering examples with the use of chiral single molecule catalysts have been reported for the anomalous *exo*-selective enantioselective Diels-Alder reaction. On the other hand, catalysts for the anomalous *endo*-selective Diels-Alder reaction have been limited in the achiral approach. To establish high catalytic activity with anomalous *endo/exo*- and enantioselectivity, in principle, an artificial cage technique is attractive since conformationally flexible structure, like an active pocket in an enzyme with an induced-fit function, should be effective. In this context, we recently developed novel supramolecular catalysts, which were readily prepared *in situ* from chiral 3,3'-disubstituted binaphthols and biphenols, arylboronic acid, and $\text{B}(\text{C}_6\text{F}_5)_3$, for promoting both the anomalous *endo*- and anomalous *exo*-selective Diels-Alder reaction. The evolution from 'ready-made' single-molecule catalysts to 'tailor-made' supramolecular catalysts could offer not only

high enantioselectivity but also high anomalous *endo/exo*-selectivities due to substrate-specific characteristics, as with enzymes. By designing conformationally flexible chiral pockets similar to those in enzymes, 'tailor-made' chiral supramolecular catalysts can be fine-tuned for each substrate to establish high anomalous substrate-selectivity and/or stereoselectivity, which is difficult to achieve with conventional 'ready-made' single-molecule catalysts.

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